

# Gabriel Ganeles

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## EDUCATION

**Technion Israel Institute of Technology**, Haifa, IL

Graduating August 2026

- Candidate for Bachelor's of Science in Biomedical Engineering, GPA: 84.9
- Relevant coursework by field:
  - Software: Digital Signal Processing, Python, Deep Learning, Seeing and Hearing Systems
  - Electrical: Electronic Circuit Theory, Semiconductors, Systems and Signals
  - Mechanical: Fatigue and Fracture Mechanics, Fluid Mechanics, CAD, Rigid Body Dynamics, Control Theory
  - Biomedical: Three Biomedical Lab Courses, Quantum Mechanics, MRI, Medical Instrumentation Design

## EXPERIENCE

**Autonomous Driving Team Leader**

August 2023 - Present

*Formula Technion – Haifa, Israel*

- Co-led a 12-member team developing the end-to-end perception, localization, control, and hardware stack for a Formula Student autonomous race car
- Designed custom circuits enabling high-speed data transfer, AI compute, and safety-critical vehicle state management
- Implemented factor-graph based SLAM, deterministic planning, and real-time system monitoring software in C
- Built the full team's public website, increasing outreach and visibility to 3,000+ monthly visitors

**Final Project for Bachelor's Degree**

September 2024 - August 2025

*Technion BioMotion Lab – Haifa, Israel*

- Automated identification of maneuvers in windsurfing video footage to reduce coach review time
- Developed a computer application to help the Israeli Olympic windsurfing team review athlete footage more efficiently

**Student Researcher**

September 2023 - July 2024

*Technion's Experimental Particle Physics Group – Haifa, Israel*

- Developed an algorithm to optimize Large Hadron Collider detector performance
- Simulated muon interactions within ATLAS' New Small Wheel to study detector overlap region activation patterns to improve muon detector (sTGC) accuracy

**Intern**

August 2023 - September 2023

*Flinker Lab at NYU Langone – New York City, NY*

- Worked with MATLAB-based data processing and medical imaging workflows
- Extracted features from electrical brain data from 50 patients to train an EEG-to-speech neural network
- Converted MRI and CT scans into 3D brain surfaces to map electrode placements into brain regions

**Mechanical Team Member**

September 2022 - July 2023

*Technion Rocketry Club – Haifa, Israel*

- Created a spring-based 3D printed parachute launcher for a hybrid solid-gas model rocket
- Collaborated across propulsion and avionics teams to develop the rocket's structural body
- Supported mechanical design and prototyping work for the club's pneumatic systems

**Robotics Engineer**

January 2021 - July 2023

*Technion Automated Robotics Lab – Haifa, IL*

- Built and programmed an arduino driven 3D printed robotic gripper for reliable handling of salmon fillets
- Refined gripper design over multiple iterations to improve pick-and-place reliability

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## INDEPENDENT PROJECTS

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**Linux Trackpad Edge Swipe Detector** / <https://crates.io/crates/trackpad-edge-swipe> May 2026

- Developed an event driven trackpad 2 finger edge swipe detector for linux
- Tech Stack: Rust, Evdev

**GyroNova** / [gyronova.gganeles.com](https://gyronova.gganeles.com) January 2026

- Programmed a web-rtc based, phone gyroscope controlled online multiplayer game
- Tech Stack: Firebase, Flutter+Flame

**Gabe's Games Corner** / [set2.gganeles.com](https://set2.gganeles.com) September 2025

- Programmed 2 multiplayer games: Set and Anagrams (Snatch)
- Tech Stack: SolidJS, Tailwind, Rust+Warp

**SET and Other Games** / [set.gganeles.com](https://set.gganeles.com) January 2023

- Programmed 4 serverless multiplayer games in Svelte: Set, Anagrams (Snatch), Memory (Concentration), and Skull
- Tech Stack: Svelte, Firebase Hosting, Firebase Realtime Database, Tailwind-css

**Shab-BOT** / [github.com/gganeles/shabbot](https://github.com/gganeles/shabbot) March 2022

- Developed in Go+Javascript, functionality includes: Shabbat meal guest tracking, location dependent Shabbat candle-lighting time, whatsapp-based reminders and scheduled messages.
- Tech Stack: Whatsmeow, Chrono-node, Oracle VM

## SKILLS

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**3D Modeling:** Creo, Onshape, SOLIDWORKS, Blender, Rhino, Grasshopper

**Electronic Circuit Design:** LTSpice, Altium, KiCad

**Programming Languages:** Rust, Go, Kotlin, C, C++, Python, MATLAB, Javascript, HTML, CSS, Linux, Bash

**Development Tools:** Svelte, React, SolidJS, ROS2, Dioxus

**Languages:** English (Native), Hebrew (Fluent)

## AWARDS

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**Technion Bio-Hack Hackathon** / First Place April 2025

- Proposed a device to assist in the rapid prevention of cutaneous infection

**Nucleate Bio-Fense Hackathon** / First Place June 2024

- One month bio-convergence hackathon sponsored by DDR&D (מפא"ד) and Teva
- Developed a solution to combat-based neck injuries

## VOLUNTEERING

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**Gazelle Valley** / Park Volunteer / JERUSALEM IL September 2019 - March 2020

- Volunteered 3 hours weekly to care for the park, planting trees, cleaning litter, and clearing reeds from the swamp
- Constructed a cage for baby turtles to protect them from predatory birds